

HR Attrition Analysis Report

A Comprehensive Analysis of Employee Attrition using Data Analytics

Prepared for: Business Stakeholders

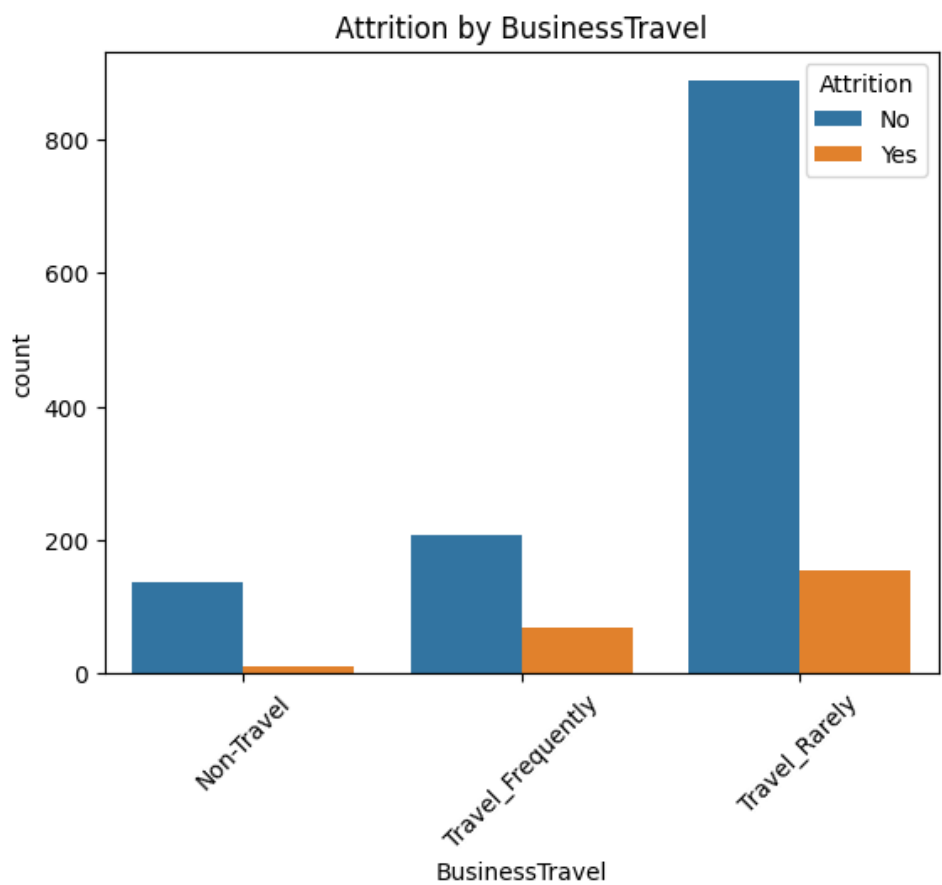
1. Introduction

This report presents a comprehensive analysis of employee attrition using structured HR data. The objective is to identify key factors influencing employee turnover and provide actionable business insights to improve retention strategies.

2. Exploratory Data Analysis (EDA)

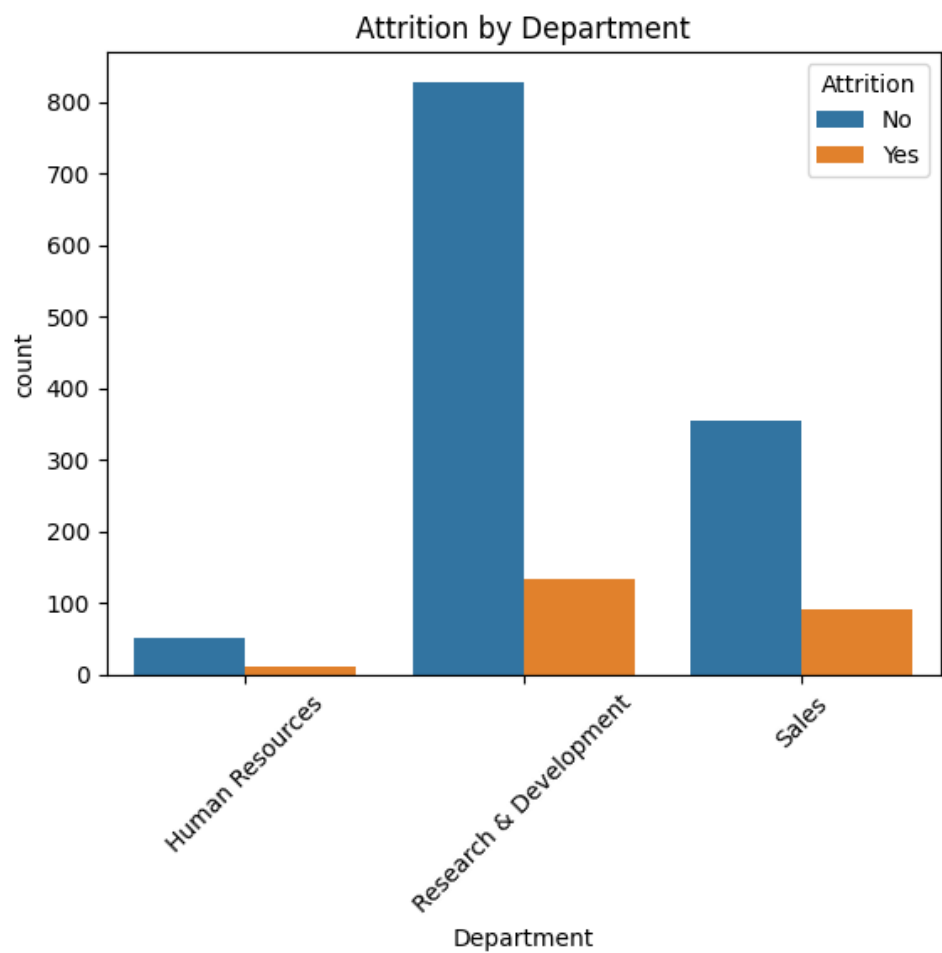
The EDA phase involved understanding distributions, detecting patterns, and identifying anomalies across multiple variables. Both univariate and bivariate analyses were conducted.

Attrition by Business Travel



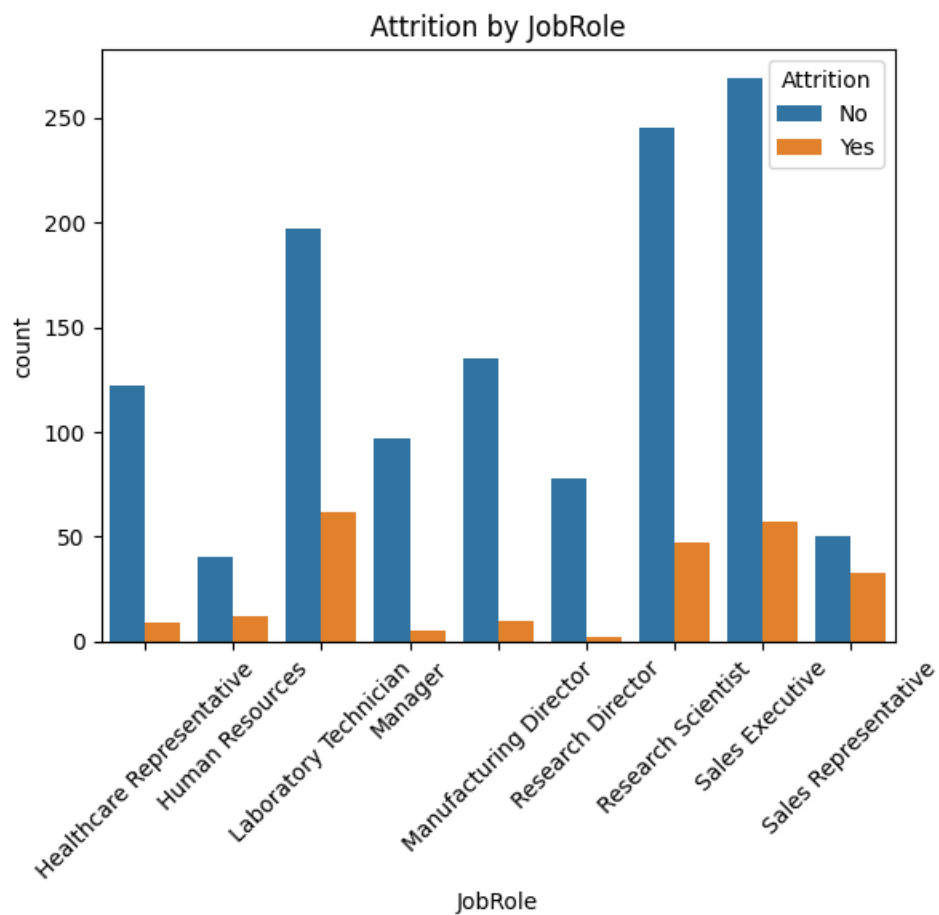
Attrition by Business Travel

Attrition by Department



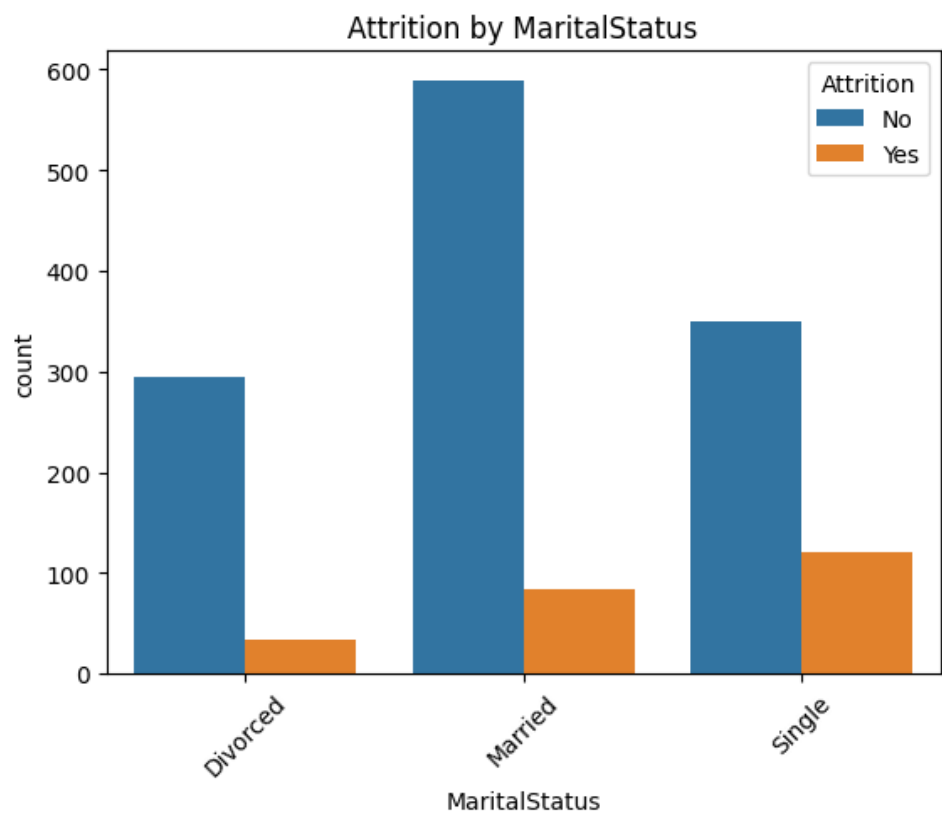
Attrition by Department

Attrition by Job Role



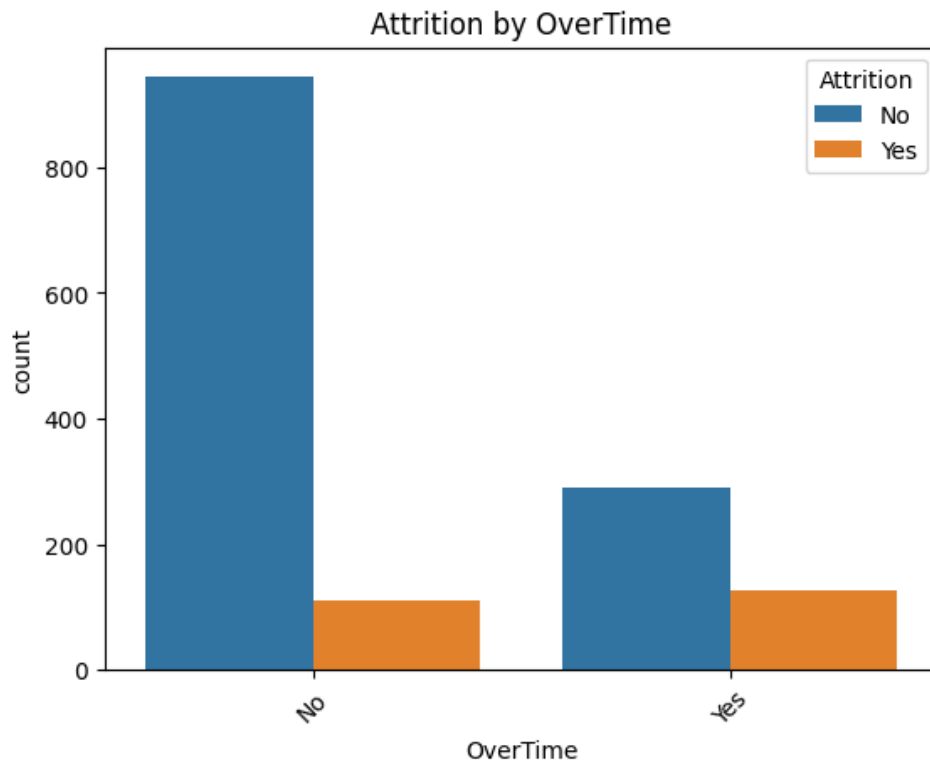
Attrition by Job Role

Attrition by Marital Status



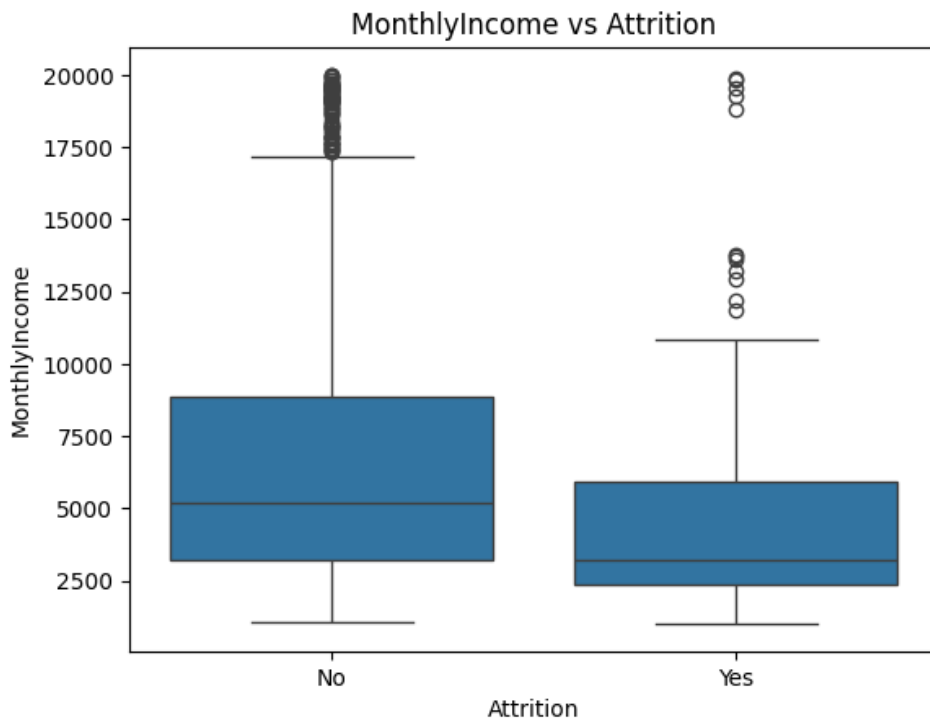
Attrition by Marital Status

Attrition by Overtime



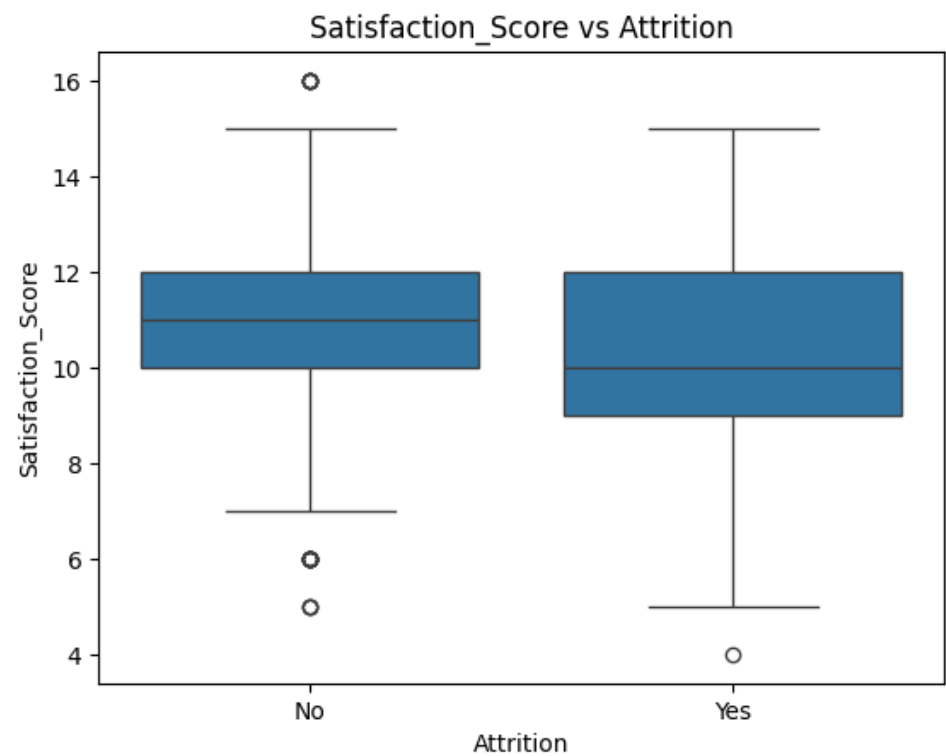
Attrition by Overtime

Monthly Income vs Attrition



Monthly Income vs Attrition

Satisfaction Score vs Attrition



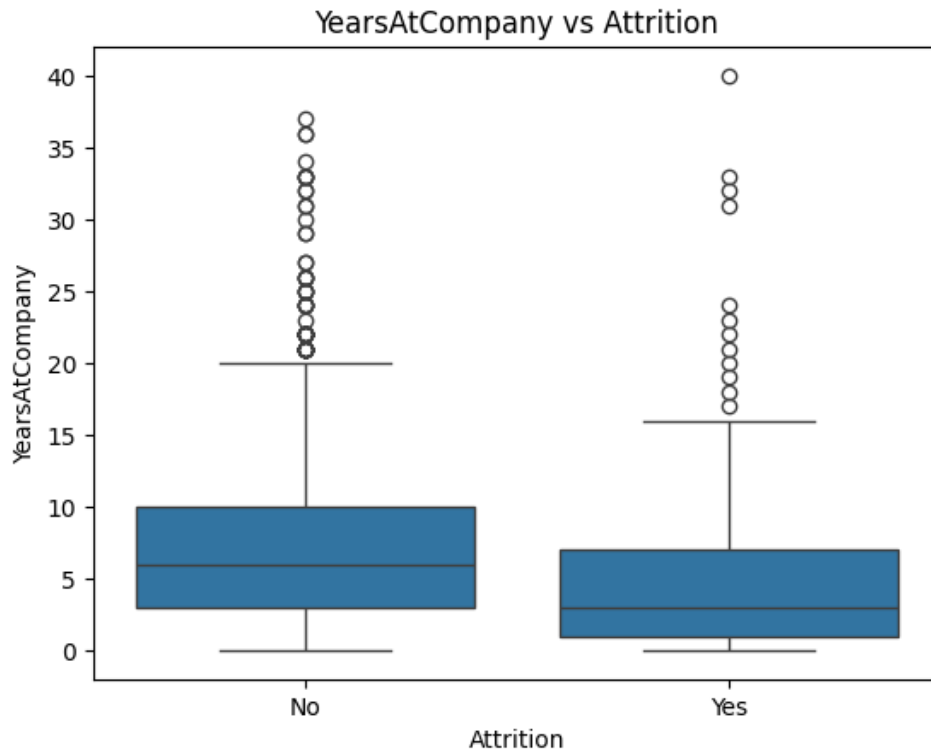
Satisfaction Score vs Attrition

Work Stress Index vs Attrition



Work Stress Index vs Attrition

Years at Company vs Attrition



Years at Company vs Attrition

3. Key Findings from EDA

- Overtime significantly increases attrition.
- Lower income employees are more likely to leave.
- Early tenure employees show higher attrition rates.
- Satisfaction levels strongly correlate with retention.
- Sales and HR departments have higher attrition.

4. Business Insights

Attrition is driven by a combination of behavioral, financial, and experiential factors. Employees with low satisfaction, high stress, low tenure, and low income are at highest risk.

5. Recommendations

- Optimize compensation structures.
- Reduce overtime and improve workload balance.
- Strengthen onboarding programs.
- Monitor employee satisfaction regularly.

- Implement predictive analytics for retention.

6. Conclusion

Attrition is predictable and manageable. With data-driven strategies, organizations can significantly reduce turnover and improve workforce stability.